

From Theory to Practice: The Sim-Etico Simulation for Ethical Military Education*

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Abstract: The challenge of developing reflective ethical judgment in military students, particularly within high-pressure operational contexts and complex contemporary scenarios, calls for a critical re-evaluation of traditional didactic strategies. This article addresses this need by presenting the design of the Sim-ETICO simulator, a novel ICT-based tool. The study is grounded in the Design Science Research paradigm, focusing on instructional architecture and pedagogical efficacy rather than empirical testing. The core contribution is Sim-ETICO, which aims to transform ethical theory into practical, scenario-based decision-making. By utilising case studies, the simulator trains the competencies required for military personnel to transition from civilians to responsible professionals committed to institutional and social values. The research methodology involved a systematic evaluation of existing ethical training strategies, followed by the selection of tentative technical and pedagogical components derived from literature reviews, semi-structured interviews, and focus groups with lecturers, students, a manager, and an ICT expert. The final design was validated using the Delphi methodology with a qualified panel of educators and military experts. Validation outcomes confirmed the strong theoretical coherence and pedagogical viability of the Sim-ETICO instructional design for fostering ethical formation. This framework provides a model for integrating technology to address the competence gap in military ethics. However, it is important to note that this article only reports on the design phases and validation panel; it does not present empirical data regarding the impact on student learning outcomes, which will be the focus of subsequent investigations.

Keywords: educational strategies; ethics; generations; ICT

* Research article.

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Resumen: Desarrollar un juicio ético reflexivo en los estudiantes militares es un desafío, particularmente en contextos de alta presión operativa y escenarios contemporáneos complejos, lo que requiere una evaluación crítica de las estrategias didácticas tradicionales empleadas en la actualidad. Este artículo aborda esta necesidad presentando el diseño del simulador Sim-Etico, una novedosa herramienta basada en TIC. El estudio se fundamenta en un enfoque de investigación basado en diseño, enfocándose en la arquitectura instruccional y la eficacia pedagógica más que en pruebas empíricas, definiendo así claramente su alcance metodológico. La principal contribución es el marco instruccional detallado de Sim-Etico, que busca transformar la teoría ética en toma de decisiones práctica basada en escenarios específicos. Mediante el uso de estudios de caso, el simulador entrena activamente las competencias requeridas para que quienes ingresan como civiles en la carrera militar se conviertan en profesionales responsables, comprometidos con los valores institucionales. La metodología de investigación incluyó una evaluación sistemática de las estrategias existentes de formación ética, seguida de la selección de componentes técnicos y pedagógicos tentativos como resultado de la revisión de literatura, entrevistas semiestructuradas y grupos focales con docentes, estudiantes, un directivo y un experto en TIC. El diseño final fue sometido a validación utilizando la metodología Delphi con un panel calificado de educadores y expertos militares. Los resultados de la validación confirmaron la fuerte coherencia teórica y viabilidad pedagógica del diseño instruccional de Sim-Etico para fomentar una formación ética activa. Este marco ofrece un modelo sólido para integrar la tecnología y abordar la brecha de competencias en ética militar. Sin embargo, es esencial señalar que este artículo solo reporta las fases de diseño y validación por expertos; no presenta datos empíricos sobre el impacto en los resultados de aprendizaje de los estudiantes, lo cual constituye la fase subsiguiente necesaria de la presente investigación.

Palabras clave: estrategias educativas; ética; taxonomía generacional; TIC

Resumo: O desafio de desenvolver o julgamento ético reflexivo em estudantes militares, especialmente em contextos de alta pressão operacional e cenários contemporâneos complexos, exige uma reavaliação crítica das estratégias didáticas tradicionais. Este artigo aborda essa necessidade ao apresentar o design do simulador Sim-Etico, uma ferramenta inovadora baseada em TIC. O estudo está fundamentado na abordagem de pesquisa baseada em design, com foco na arquitetura instrucional e na eficácia pedagógica, em vez de testes empíricos, definindo assim claramente seu escopo metodológico. A principal contribuição é o detalhado marco instrucional do Sim-Etico, que busca transformar a teoria ética em tomada de decisão prática baseada em cenários. Utilizando estudos de caso, o simulador treina ativamente as competências necessárias para que o pessoal militar faça a transição de civis para profissionais responsáveis, comprometidos com valores institucionais e sociais. A metodologia de pesquisa envolveu uma avaliação sistemática das estratégias existentes de formação ética, seguida da seleção de componentes técnicos e pedagógicos tentativos como resultado de revisão de literatura, entrevistas semiestructuradas e grupos focais com docentes, alunos, um gestor e um especialista em TIC. O design final foi submetido à validação utilizando a metodologia Delphi com um painel qualificado de educadores e especialistas militares. Os resultados da validação confirmaram a forte coerência teórica e a viabilidade pedagógica do design instrucional do Sim-Etico para promover uma formação ética ativa. Esse marco oferece um modelo robusto para integrar a tecnologia e enfrentar a lacuna de competências em ética militar. No entanto, é essencial ressaltar que este artigo apenas relata as fases de design e validação por especialistas; não apresenta dados empíricos sobre o impacto nos resultados de aprendizagem dos estudantes, o que constitui a próxima fase necessária de investigação.

Palavras-chave: estratégias educacionais; ética; taxonomía geracional; TIC

Introduction

Teaching in military contexts is a challenging process because it must integrate the demands of military traditions, the specific field of knowledge (e.g., aeronautics), and the broader field of education. Each of these elements has its own policies, practices, and objectives, creating a unique teaching-learning model. One example is the slow pace of implementing innovative educational practices, which often results in outdated habits, despite significant advances in both generations and technology over the past 50 years.

The need to instruct new military learners in ethical conduct is critical, as their actions must be beyond reproach and dedicated to defending the life, honour, and property of all citizens. Military ethics are universally applicable, regardless of branch or nation, and improper application can have severe consequences, including crimes against humanity. Thinkers, from classical philosophers to contemporary authors like Habermas, have emphasized that ethical reflection should not be confined to theoretical instruction alone but must include opportunities for dialogue and discussion. While ethical principles can be conveyed through lectures, the development of ethical reasoning requires pedagogical approaches that foster interaction and debate among students.

An example of such a pedagogical approach is the NCO School of the Colombian Aerospace Force, one of the three academies within this military branch. Based in Madrid, Cundinamarca, Colombia, it annually trains young men and women from across the country, with students aged between 16 and 21 for regular courses, and up to 28 years old for administrative and extraordinary courses. All students engage with an ethics module during their first three months of training, focusing on the values, principles, and virtues established by the institution, which are outlined in a manual. While this serves as an initial introduction to the topic, it is crucial for the transformation from civilian to military life.

Over recent years, several issues have been identified within this context, which can be summarized at three levels:

Micro-context: The grades achieved in this module are often unremarkable, with many students only reaching the minimum passing score, despite the module's significant relevance

Meso-context: The Ministry of Education requires the TandT test for granting the technological title. However, results in areas directly related to ethics often fall behind those of civilian career paths, even though the military profession involves handling weapons and requires behavior free from reproach.

Macro-context: A significant number of disciplinary, administrative, and criminal investigations are initiated in the year following technological graduation, when ethical behavior should be expected due to the recent knowledge acquired.

These challenges lead to the research question: Which didactic strategy could enhance the learning of the military ethics module at the NCO School of the Colombian Aerospace Force? To address this question, a comprehensive review of sources was conducted. Literature from databases such as Scopus, EBSCO, and ERIC, including books, articles, and research from 2018 onwards, reveals considerable interest in the study's topics. However, no single source comprehensively addresses all aspects, highlighting a gap in the existing literature. Notably, when exploring the relationship between ethics, educational strategies, and ICTs across various regions, it is clear that ethics serves as a central axis of connection between pedagogical approaches, ICTs, and the digital skills required for action in virtual environments, a trend developing globally.

Building on the unique characteristics of the chosen context, this research sets three objectives: first, to diagnose the current strategy for teaching military ethics; second, to identify the technical elements needed for proposing an alternative strategy; and third, to validate that proposal. Achieving these objectives starts with a literature review of key theories from different countries, focusing on thematic areas such as ethics, ICT in education, and generational traits.

All objectives ultimately converge in the proposal of an ethical simulator, drawing inspiration from pilot simulators and air control tower simulators. The proposed SimÉTICO (in Spanish) aims

to present real ethical dilemmas in a controlled environment, without real-world consequences. The proposal is based on Problem-Based Learning (PBL) and Salmon's E-Learning Interactivity Model.

It is important to clarify the limitations of this study. The research does not attempt to cover all possible scenarios that could arise in combat but focuses on situations directly related to the fundamental behaviours expected of every new member of the institution. Therefore, the proposal includes only a limited number of simple, yet realistic situations encountered after graduation. Other subjects, such as Human Rights or International Humanitarian Law, delve into more complex issues, such as those experienced in armed conflict. However, this study aims to take the first step in the teaching and learning process, focusing on the basics before advancing to more specialized fields of knowledge. Additionally, while military ethics includes topics like artificial intelligence and cyber operations, this research specifically concentrates on ethical decision-making at the tactical and operational levels, excluding scenarios involving autonomous systems. The integration of AI-based simulations is reserved for future work and is not addressed in this study.

Given these constraints, it was necessary to focus on core objectives and methodologies that could be realistically achieved within the scope of the project. However, AI represents a promising area for future exploration and could be considered in a subsequent phase of research, potentially expanding the impact and capabilities of the proposed didactic strategy.

Finally, this article is organized into five main sections: 1. Literature Review with the exposition of

the theoretical framework and the current state of the investigations; 2. Methodology, with the explanation of the decisions taken to ensure a clear and detailed process; 3. Results structured in 2 areas (the outcomes in every objective, and the proposed intervention); 4. Discussion examines the findings in relation to the research question and the broader literature; 5. Conclusion summarises the main contributions of the study, highlighting the innovative aspects and potential impact of the proposed didactic strategy.

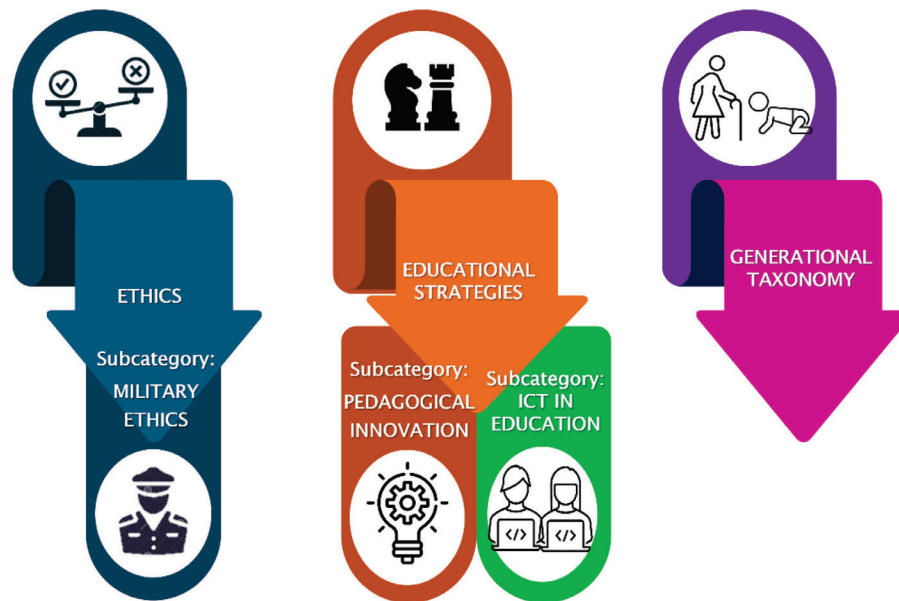
Literature Review

This section presents the literature review, divided into three parts, to outline the theoretical framework and current perspective on the key categories identified as central to addressing the research question: Which didactic strategy could enhance the learning of the military ethics module at the NCO School of the Colombian Aerospace Force? The categories explored are ethics, educational strategies, and generational taxonomy.

Ethics is the primary focus of the teaching and learning process.

Educational strategies are examined because the context has been using the same types of tools for decades, with little evolution. Generational taxonomy is considered because the current students have distinct motivations, interests, and characteristics that need to be incorporated into any proposed solution. Each category begins with a review of foundational theoretical concepts, followed by a concise summary of the current state of research in that area.

Figure 1. Theoretical categories



Source: Author.

Ethics

Ethics is a universal concept that continually evolves, tracing its roots from ancient philosophical reflections (e.g., Socrates, Aristotle) to contemporary debates. Historically, ethical thought can be broadly divided into two main currents: a humanistic approach, which focuses on inner conviction and “the good,” and an authoritarian approach, which imposes external order on society (Fromm, 2016). Contemporary discourse recognizes that while values are rooted in specific historical and communal contexts, the idea of a universal ethics requires ongoing dialogue and reflection among humanity (Habermas, 2003).

A critical modern perspective is provided by Adela Cortina (2013a), who distinguishes between “ethics of maximums” (personal, lived experiences) and “ethics of minimums” (socially accepted values). The latter is crucial as it emphasizes the norms required for peaceful social coexistence, allowing the collective good to prevail over individual desires for the sake of species transcendence. Fundamentally, ethics represents an “internal transformation,” driven by the conviction that doing good is inherently valuable (Cortina, 2013b,

p. 45). This internal shift directly influences daily decision-making.

This philosophical foundation translates into a pedagogical imperative within higher education. Current research asserts that ethical instruction is essential for cultivating responsible, conscious citizens capable of making reflective and fair decisions in a complex world (Ziadet Bermúdez et al., 2025). While traditional methods, such as lectures, remain prevalent (Bello and Colmenares, 2015), there is a growing consensus that they are insufficient. Contemporary challenges demand a focus on ethical critical thinking, digital ethics, and practical application through methods like case studies and mentoring to effectively prepare professionals for their careers (Pérez Perdomo et al., 2024). This highlights the pressing need to shift from purely theoretical knowledge to comprehensive ethical competency training.

The foundation of military ethics can be traced back to Clausewitz (1873), who established that military virtue should serve as the guiding principle in situations where direct command is absent. The application of these virtues must be so deeply ingrained that service members can exercise effective self-leadership, aligned with institutional

ethics, even in the absence of direct supervision. This imperative is starkly illustrated by Arendt's (2013) study of the "banality of evil," which concluded that strict adherence to orders—such as in the Eichmann case—does not absolve military professionals of their "inability to think" or make ethical decisions (p. 35). Thus, military personnel must be trained to make ethical judgments, even when orders appear to contradict their moral compass, emphasizing the critical role of ethical leadership at every rank.

The military context introduces a unique tension: the institutional mission demands esprit de corps and sacrifice, often creating a conflict between individual conscience and group values (Moskos, 1984; Sarkesian, 2013). While foundational theorists like Huntington (1981) advocated for the separation of the military from civilian society, this research aligns more closely with Janowitz (1990) and Gómez (2017), asserting that professional military ethics must reflect the moral codes that govern all human beings, as soldiers originate from and serve society. Ultimately, ethics supports the soldier in the proper execution of their duties, even when applying force, while reminding them that relinquishing personal autonomy in service does not absolve them of responsibility for their actions (Gómez, 2017; Moliner, 2020). Despite the consensus on the importance of comprehensive ethical training, traditional methodologies in military contexts remain largely focused on lectures (Aponte López, 2024; Villareal, 2013), a method often considered inefficient for teaching ethics (Castellanos, 2014). Effective alternatives—such as war games, technological platforms (Pinto, 2017), and case studies—share the common objective of stimulating critical thinking and decision-making, rather than merely promoting rote memorization (Aponte López, 2024; Villareal, 2013). This need for innovative approaches is exemplified by leaders like General Rick Lynch in Iraq (Quick, 2025), who successfully used commercial films and discussion-based training to significantly reduce ethical transgressions. This collective evidence confirms that ethical training is not simply another class; it requires practical, cross-disciplinary activities using tools that prepare students for professional

life by creating behavioural antecedents in a controlled, safe environment.

Educational Strategies

The field of educational strategy, which has been reflecting on the "art and ability of teaching" since Comenius (1998), has long emphasized the importance of adapting methods to meet student needs (Montessori, 1974) and incorporating action, application, and exercise (Aebli, 2001). Modern definitions formalize this concept, viewing strategy as a flexible "action plan" (Tobón, 2010) or a selection of pedagogical activities (Velasco and Mosquera, 2010) specifically designed to address contextualized problems (Monereo, 2005). Crucially, effective strategies must "facilitate the confrontation (interactivity)" of the learner with the object of knowledge (Ferreiro, 2012).

This foundational shift towards action-based learning has led to the widespread adoption of Problem-Based Learning (PBL), advocated by theorists such as Bruner and Dewey (Gómez, 2005). PBL is a vital educational tool because it centers on the student, presenting genuine, complex situations that significantly enhance conceptual, procedural, and attitudinal competencies (Hernández and Yallico, 2020). Furthermore, educational strategies must be mediated by available technologies and cater to students' individual learning preferences (Campos, 2003; Rojas-Palacio, 2024). The effectiveness of PBL is notably amplified when motivational elements like gamification and virtual learning are integrated (Martínez, 2021; Mejía, 2021). Ultimately, effective didactic design requires careful consideration of practice, context, and students' interests to foster meaningful relationships with both knowledge and peers.

Pedagogical Innovation

Pedagogical innovation is defined as the deliberate and structured introduction of new teaching methods that challenge established practices, with the goal of enhancing learning outcomes and fostering deeper understanding (Walder, 2017; González and Cruzat, 2019). This vital shift promotes learning that is intrinsically enjoyable

(Csikszentmihalyi, 1990) and requires holistic institutional renewal, including the professional development of teachers and the integration of ICT for curriculum development (Escudero, 1984). The core challenge is adaptation: ensuring that the teaching process is responsive to the evolving environment and the specific needs of students (Montessori, 1974).

This imperative for change is driven by current technological advancements, which have led to new models such as Connectivism (Siemens, 2010). Connectivism integrates network-based learning and the concept of learning ecology and is essential because the digital generation exhibits significant changes in memory and attention, showing a preference for active, screen-based, and multimodal learning experiences over passive, traditional methods (Mynbayeva et al., 2018; Lonka, 2018). Effective innovation must, therefore, strike a balance—aligning new tools with core educational traditions to develop essential competencies like communication skills and long-term memory (Mynbayeva et al., 2018).

As a result, successful pedagogical strategies today often incorporate motivational elements such as gamification, which applies game design elements in non-game contexts (Deterding et al., 2011). In virtual environments, this fosters critical engagement through avatars, narratives, and feedback. Ultimately, this disruptive form of experiential learning facilitates the development of essential soft skills—such as critical thinking and resilience—by allowing students to apply knowledge in realistic scenarios (Escamilla, 2019; Nguyen et al., 2025).

ICT in Education

The challenge for modern education is clear: technological change is “irreversible” (Postman, 1975, p. 17). If institutions fail to align with the societal shift toward the Information Age, they risk generating a “shock of the future,” leaving students unprepared for the world beyond the classroom (Postman, 1975). This urgency is heightened by the fact that knowledge, now transmitted through networks in what Echeverri (2015) refers to as the “Third

Environment,” has become the most vital resource (Stewart, 2010; Drucker, 2012). As a result, curricula must evolve beyond the constraints of traditional methods, focusing on “knowledge workers” and developing skills that enable the management and application of information (Drucker, 2012).

Effectively integrating ICT requires a thoughtful, staged approach, as outlined in Salmon’s E-learning model (2013), which emphasizes a progression from initial motivation to the final construction of knowledge. This digital shift makes learning increasingly lifelong, wireless, and decoupled from physical locations (Coll, 2013), highlighting the need for transversal competencies. For example, the successful use of computational models to teach complex concepts demonstrates the power of transforming research tools into educational resources (Tinker, 2008). Therefore, ICT should not merely be seen as a tool, but as a crucial “learning vehicle” (Coll, 2013). This research aligns with the consensus that integrating ICT into military training, in particular, requires a redefinition of traditional pedagogical approaches and a greater investment in fostering innovation (Escrura and Canese, 2024). Ultimately, while ICT does not directly create skills, it motivates their development by providing an active, enabling environment for tools like Sim-ETICO (Moreno, 2016).

Generational Taxonomy

The concept of distinct generations is rooted in the historicity of worldviews (Dilthey, 1961), shaped by shared physical contexts, values, and defining events (Mannheim, 1923). This collective experience fosters a unique “vital sensibility,” creating bonds among peers while often generating tensions with the preceding cohort (Ortega y Gasset, 1934)—a dynamic clearly observable in contemporary military and higher education settings.

This generational conflict is frequently framed through the lens of digital taxonomy, distinguishing between older generations—often categorized as ‘Digital Visitors’—and current students, identified as ‘Digital Natives’ or Generation Z (Prensky, 2015; Pew Research Center). Millennials (Howe and Strauss, 2009) and especially Gen Z (born

from 1997 onwards) are immersed in the digital ecosystem from birth. They exhibit unique learning preferences, including a demand for immediate access to information, a need for constant feedback, and a tendency to thrive on multitasking and screen-based communication (Cárdenas and Cáceres, 2019; Lonka, 2018).

This fundamental shift creates a significant pedagogical gap. Traditional educational strategies, often favored by lecturers from older generations, may alienate students who are accustomed to the interactive, connected nature of online games and social media (Sanguino, 2021). As Gen Z's learning styles continue to evolve, adopting new approaches that align with their preferences and leverage the latest technological advancements is essential (Challa et al., 2024; Guzman, 2024). Therefore, training processes must consciously integrate these generational characteristics, ensuring that curricula are designed and developed to meet the immediate and digital needs of today's students.

Methodology

Research Methodology and Data Collection Process

This research follows the approach outlined by Hernández et al. (2014), beginning with the specific and moving towards the general. It navigates between facts and their interpretation, culminating in an inductive process aimed at understanding the socio-educational reality of a higher education institution with a technological focus, specifically within a military context.

The choice of this approach is justified by the need to identify the specific characteristics of the military educational environment, which is shaped by unique norms and traditions. These elements leave a distinct mark on the training process, particularly in military ethics. Moreover, including the experiences and perceptions of both students and lecturers is seen as valuable input for developing a proposal that can improve teaching and learning conditions within this institution.

Data Collection Methods

Focus Groups

Focus groups were selected as the primary data collection method for addressing the first research objective. This method considers the group as the unit of analysis, emphasizing group dynamics and collective insights (Hernández et al., 2014). The participant selection process followed Creswell's (2005) recommendations: for complex issues like military ethics, focus groups with six to ten participants are optimal.

The initial diagnostic phase targeted a non-probabilistic sample from the NCO School of the Colombian Aerospace Force. Two focus groups were conducted—one with ten final-year students and another with ten lecturers. In addition, two semi-structured interviews were carried out: one with an institutional manager and one with an external expert in virtual environments. The focus group discussions and interviews were structured around five core thematic categories: Preconceptions before joining the institution, Pedagogy, Ethics, Generational taxonomy, ICTs

Each focus group session was guided by 14-16 open-ended questions designed to explore real-life ethical dilemmas.

Semi-Structured Interviews

A second data collection tool was employed: semi-structured interviews. The first interview was with an institutional manager responsible for overseeing all technological programs at the institution. This individual's extensive knowledge and experience (15+ years in educational processes) made them a key informant for the study. This interview contained 9 questions focusing on the technological aspects of military education.

The second semi-structured interview was conducted with a faculty member from Pontificia Universidad Javeriana, an expert in virtual reality, augmented reality, and digital environments. The interview, with ten questions, addressed four main topics: the current relevance of the proposal,

compliance with the research objective, feasibility of implementation, and the potential for future development. Their insights were crucial for assessing the feasibility and potential of integrating new technologies in military education.

Validation Process

To ensure the proposed pedagogical intervention was relevant and feasible, it was validated through expert consensus using the Delphi method. This process gathered feedback from five specialists with at least six years of experience at the institution, all holding a master's degree and actively participating in academic innovation initiatives. The panel validated the data collection tools (interviews and focus groups) and the final pedagogical proposal.

In addition to primary data sources, secondary sources such as the master documents of each academic program were also used to supplement the research.

Data Analysis

For analysing the collected data, the Content Analysis model was applied. This qualitative approach, as defined by Andréu (2002), interprets various forms of text (written, recorded, visual) by systematically examining their content to gain insights into social phenomena. Content Analysis is particularly effective in combining data with contextual interpretation, enabling a deeper understanding of the subject matter.

Given the study's objectives, Content Analysis was chosen for its structured process, which helps organize, categorize, and analyze information efficiently. The analysis began with transcribing the discussions and opinions generated during focus group and interview sessions. Accurate transcription was crucial to maintain the integrity of the data and avoid misrepresentation, especially considering the use of colloquial language and military-specific terminology (Gibbs, 2012).

To support the analysis, the nvivo application was employed, leveraging artificial intelligence for language recognition and interpretation. Manual verification of transcriptions was performed

by comparing audio recordings with written transcripts to ensure accuracy, particularly regarding colloquial and technical language.

The analysis process proceeded through three main stages:

1. **Data Reduction:** Information was broken down into smaller units for easier analysis. Codes or labels were assigned to these units, often using colors for quick identification.
2. **Data Organization:** The codes were grouped into categories based on common themes such as opinions, situations, and activities identified in the focus groups and interviews.
3. **Results Generation:** The categorized data were analyzed to draw conclusions and generate insights, which were presented in the results chapter.

Once data reduction was completed, the information was organized by topics and theoretical concepts, with relationships established to draw overall conclusions. All collected data were deemed valuable for providing both a broad and focused view of the studied reality, and no additional instruments were necessary to fill gaps.

Results

This section presents the findings derived from the data collection tools discussed earlier, including interviews and focus groups. These results will be further analyzed in Section 4.

The core research question guiding the data collection was: What ICT-based didactic strategy could strengthen the teaching and learning of ethics in a military higher education institution? The analysis of the data from the first focus group with final-year students at the chosen Military Training School was conducted using content analysis, which led to the identification of several key categories.

Student Focus Group Findings

The data from the student focus group were organized into a hierarchical structure, initially identifying 21 distinct categories. These were

subsequently condensed through content analysis. As seen in Figure 2, prominent themes included specialty, school, military, example, beliefs, motivation, and courtesy.

A key observation was that the current pedagogical strategy heavily relies on lectures and theoretical content, while students expressed a strong interest in more interactive, student-centered approaches.

Students also emphasized the operational necessity of practical ethical training. A notable frustration expressed was the lack of realism in current instruction, particularly the gap between theoretical knowledge and its application in real-world military situations. One final-year student remarked: “If you can focus and view everything from a realistic point of view, you can keep practicing the information. It’s not just sitting in class and listening to lectures” (translated by author).

This data underscores the need for practical, experience-based learning. The student focus group identified the use of a simulator as not just an educational preference, but a critical requirement for developing practical knowledge and ethical decision-making skills.

Lecturer Focus Group Findings

The second focus group, consisting of lecturers, followed a similar structure to the student group. The main themes identified by the lecturers were military and good examples, reflecting the importance of role models in military training. This finding aligned with students’ responses, although additional topics such as training, process, the person, and the metaverse were also highlighted.

The concept of “example” emerged as a central theme, stressing the importance of instructors and commanders demonstrating ethical behavior. This aligns with contemporary educational theories that emphasize the role of lived experience and social learning in ethics education.

Lecturers also reinforced the need for active, participatory learning methods, particularly when engaging Generation Z students. One lecturer observed: “Years go by, and generations change, but the same program persists. We can’t teach today’s

students the same way we were taught. Children today have decision-making power at age 5, which we didn’t have at 16” (translated by author).

This finding supports the argument that traditional, lecture-based methods are ineffective in developing the complex ethical competencies required in modern military contexts. It emphasizes the need for immersive, technology-driven pedagogical tools like Sim-ETICO.

Both students and lecturers highlighted the importance of motivation in the learning process. Practical activities and ICT tools were identified as crucial for enhancing engagement and helping students internalize ethical principles.

Semi-Structured Interview Findings

Two semi-structured interviews were conducted, and the data from these were also subjected to content analysis.

Manager Interview

The first interview was with an institutional manager responsible for overseeing academic programs. This manager’s perspective aligned with the findings from the student focus group regarding the importance of motivation and military themes. The manager also echoed the lecturers’ concerns about the need for continuous training and the integration of new technologies, such as the metaverse and virtual reality, into the curriculum to modernize pedagogical strategies and better prepare students for real-world ethical challenges.

One key finding from the manager’s interview was the recognition of a gap between theoretical understanding and the practical development of ethical judgment. The manager noted: “Ethics is simply not a tangible concept... you only truly understand whether a student has internalized it when they face situations that demand a high level of ethical behavior” (translated by author). This statement further validates the need for tools like Sim-ETICO that allow the observation and assessment of ethical decision-making under simulated pressure.

Expert Interview

The second interview was with an expert in virtual reality and immersive educational technologies. This expert's insights were invaluable in confirming the feasibility of using immersive technologies to enhance ethical decision-making skills in a controlled environment. Key themes that emerged from this interview included virtual reality, platforms, experience, and technology. The expert cautioned that while immersive technologies hold enormous potential, they must be carefully integrated into the educational process.

The expert emphasized that many projects focus too much on the technology itself rather than its educational value. As they explained: "In my personal experience, I've seen immense potential in immersive technologies, but they are often misinterpreted. Some projects focus solely on the technology, rather than on what it can contribute to the educational process" (translated by author). This feedback was instrumental in guiding the design of the proposed pedagogical tool, ensuring that instructional design remains the priority.

Conclusions and Implications

The findings from both the focus groups and interviews suggest a clear need for a shift in

pedagogical strategy in the military higher education context. Both students and lecturers agree that traditional, lecture-based methods are insufficient for developing the complex ethical competencies required in modern military contexts. There is strong support for the integration of ICT-based tools, such as Sim-ETICO, which can offer realistic, immersive experiences for ethical decision-making.

Moreover, the interviews with the institutional manager and the expert in immersive technologies confirmed the technological feasibility of these tools, while also highlighting the importance of focusing on pedagogical goals rather than technological sophistication.

These findings align with contemporary educational theories that advocate for active, experiential learning and the use of digital technologies to meet the needs of new generations of learners. Therefore, effective ethics education in military contexts requires a move from traditional, lecture-centered methods to innovative, technology-enhanced strategies that foster motivation, practical application, and reflective practice.

In conclusion, the current pedagogical strategy used at the institution can be summarized in the figure 2.

Figure 2. Characteristics of the current pedagogical strategy



Source: Author.

Current Teaching Practices and the Need for Innovation

Lectures remain the cornerstone of teaching and learning in most subjects, including military ethics. Theoretical content predominates, leaving little room for practical application, and the educational dynamics are largely lecturer centered. While there are well-structured academic programs in place, an update is needed to incorporate elements that foster greater student motivation. Many lecturers have not yet embraced such innovations, which, from the students' perspective, could significantly improve academic performance.

An additional challenge is the varying levels of technological adoption among lecturers. While some instructors are well-equipped to use technological resources effectively, others still rely on traditional methods like slides, limiting the scope of student engagement. Furthermore, despite their expertise in their field, not all lecturers adapt their teaching strategies to align with the specific generational characteristics of today's students.

Another factor influencing the current pedagogical strategy is the heavy reliance on standardized doctrine. This unified approach, codified through various institutional documents, ensures a sense of identity and continuity in military education. However, this standardization also risks overlooking the need for innovation in how military ethics are taught. While doctrine remains essential, the rigid adherence to it without integrating modern pedagogical strategies may stifle student engagement and reduce learning outcomes.

ICT-Based Didactic Proposal for Military Ethics

Based on the analysis above, a new ICT-based didactic approach for teaching military ethics has been proposed. It is crucial that the doctrinal aspects of military education, including the Code of Ethics, remain central to this new proposal. However, the role of the student must be emphasized alongside that of the lecturer, fostering a more interactive and student-centered learning environment. This approach seeks to enhance motivation

through gamification and the use of recognition as external motivation, encouraging active participation and engagement with the material.

The current teaching strategy, which focuses heavily on theoretical components, needs to incorporate practical elements to facilitate real-world application. In this context, the use of ICT tools and the metaverse emerges as a promising solution. These technologies offer platforms that present information in a more immersive and engaging way, akin to the experience of a flight simulator, but for ethical decision-making. This ethical simulator would allow students to practice and internalize ethical principles in a risk-free, controlled environment.

Data Collection and Analysis

The data gathered through interviews, focus groups, and a review of the literature were systematically analyzed and reduced into key categories. This process revealed the primary practices in place within the institution, as perceived by both students and lecturers. Eight key characteristics emerged from the analysis, shedding light on the strengths and weaknesses of current approaches. These insights provided the foundation for developing a solution to the central research question: What didactic strategy could enhance the teaching of military ethics at the NCO School of the Colombian Aerospace Force?

Based on these findings, the most appropriate technical components were selected to form a new proposal, which aims to address the gaps in the current strategy and better align with the needs of the educational context. These insights and proposed strategies will be further discussed in the following section.

Proposed Intervention

Building on the previously outlined conceptual framework and contextual information, pedagogy was chosen as the foundation for this proposal, specifically using Problem-Based Learning (PBL). The proposed solution, named Sim-Etico, is an ethical simulator that integrates the principles of PBL to enhance the learning of military ethics.

Sim-Etico is designed to engage students by providing access to the core doctrinal document, the Code of Military Ethics. This foundational text serves as the basis for the content. Students are then invited to apply their knowledge in simulated,

real-life military scenarios. By engaging in these immersive situations, students can demonstrate their ethical decision-making skills, bridging the gap between theoretical understanding and practical application.

Figure 3. “Sim-ETICO”



Source: Author.

Just as a pilot does not immediately board an aircraft on their first day of training but instead spends countless hours in a simulator to learn how to react to various scenarios without risking their life or the integrity of the equipment, Sim-Etico aims to serve as the ethical simulator for students learning military ethics. In this controlled, consequence-free environment, students can practice and explore different situations, decisions, and their outcomes, all within the framework of existing regulations.

The primary objective of Sim-Etico is to foster the learning of military ethics through Problem-Based Learning (PBL) by simulating real-world scenarios within the Metaverse. This virtual space, a term first coined by Neal Stephenson in his 1992 novel *Snow Crash*, refers to a collective, immersive digital universe. Increasingly, various fields—including business, entertainment, and education—are harnessing the power of virtual environments to enhance learning experiences. Consequently,

using the Metaverse for educational purposes, especially to simulate ethical decision-making, is both logical and promising.

The scope of Sim-Etico is initially focused on the students of the Non-Commissioned Officer (NCO) School of the Colombian Aerospace Force, carefully considering their generational characteristics—such as a preference for technological tools, screen-based interaction, and hands-on learning—and their academic context, including their familiarity with military vocabulary, doctrine, uniforms, and common situations.

Core Content of Sim-Etico

The Sim-Etico proposal draws on the current military code of ethics, which includes:

- Core Values for Colombian Aerospace Force Personnel:
- Honour (and its associated behaviours: righteousness and trust),

- Value (and its associated behaviours: strength and discipline),
- Commitment (and its associated behaviours: pride and loyalty)

General Principles of Action: Integrity, Safety, Virtues that Ennoble Action: Mysticism, Justice, Temperance, Fortitude, Prudence.

By immersing students in these core values, principles, and virtues, Sim-Etico aims to help internalise them, ensuring that all members of the Force are equipped to apply them in any situation. This focus is integral to the development of ethical leadership and decision-making, both inside and outside the classroom.

How Sim-Etico Works

The simulation can be accessed on various devices, such as computers, tablets, or smartphones. Upon entering, students first watch a general introduction video, which outlines the purpose and structure of the learning process. A second video summarises the key virtues of the Colombian Aerospace Force, after which the participant embarks on their first mission.

Before starting each mission, participants are given the option to consult the Code of Military Ethics for more in-depth information, catering to those who need further clarification or prefer a more detailed visual reference. For those who feel confident in their understanding, they can proceed directly to the mission.

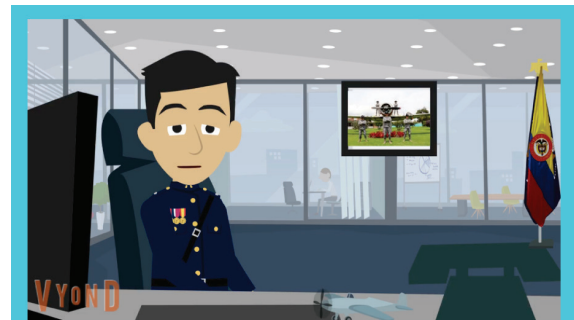
Conclusion

Sim-Etico represents a significant leap in the way military ethics are taught, providing an interactive, immersive, and technology-driven approach to learning. By integrating the Metaverse, this tool brings ethics to life in a way that engages students, enhances motivation, and bridges the gap between theory and practice. It not only provides an opportunity to learn about military ethics but also allows students to apply them in real-world-like scenarios, thereby preparing them for future challenges.

Characters. For this proposal, six distinct avatars have been created, each with unique physical characteristics that reflect the diversity of the selected institution's environment. These characters are designed to wear both the official uniforms of the institution and casual civilian clothing, as students in real life must navigate various situations requiring a wide range of attire. This design choice ensures that students can relate to their avatars, as they can select one that aligns with their personal preferences when engaging with the tasks in Sim-Etico.

Moreover, the character named CT. Andrés M. Díaz plays a central role in the simulation. As a tribute to the institution's name, CT. Díaz serves as the primary guide and evaluator within the Sim-Etico Metaverse. He leads the students throughout their virtual learning journey, offering valuable guidance and information. As host, CT. Díaz introduces each task, helps students navigate the contents of the Code of Military Ethics, and evaluates their performance in different ethical scenarios. His presence ensures that students receive continuous support, reinforcing their understanding of military ethics and providing feedback based on their decisions.

Figure 4. Main character of "Sim-Etico"



Source: Author.

Therefore, students have the option to make decisions from the very beginning by selecting their avatars, each of which represents a different military rank. Additionally, students have the flexibility to change their selected character at any point during the simulation, allowing for a personalized learning experience.

Tasks. Currently, “SimÉTICO” is an initial version consisting of three tasks or scenarios, where students will make decisions and experience specific consequences both for themselves and for the institution. If a student makes a decision that contradicts institutional ethics, *tc.* Andrés M. Díaz will prompt them to revisit the governing document, review the ethical guidelines, and attempt the task again.

On the other hand, if the student successfully applies their ethical knowledge, they will be able to see the positive outcome of their actions, receive recognition, and be promoted to a higher military rank.

The first task emphasizes institutional virtues such as justice, temperance, mysticism, fortitude, and prudence. It presents a dilemma: on one hand, the student must decide between fulfilling their professional duties by reporting punctually at 7 a.m. on a Saturday for a 24-hour security service, and on the other, the temptation of attending a party the night before with friends, where alcohol consumption and other distractions might arise.

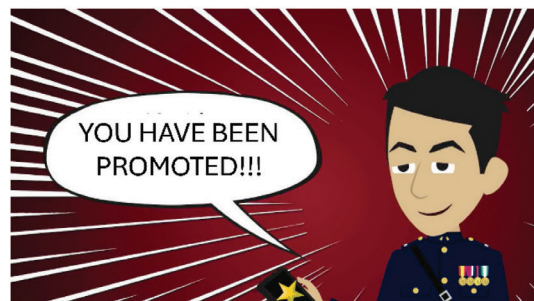
The second task emphasizes the institutional values of honor, courage, and commitment, allowing students to make decisions regarding the mandatory annual English exams for all military personnel. This task challenges the student to confront the issue of cheating, testing how they would respond based on their ethical learning.

The third task revolves around the correct way to fulfill the responsibilities of a contract supervisor—an important role that all members of the Force will take on once they graduate. This task is framed around the institutional principles of integrity and security.

To construct the full network of scenarios, 73 possible video options have been integrated so far, each with varying durations, secondary characters, and a broad range of decision-making possibilities, reflecting real-life choices and their consequences.

For the proposal to be engaging, logical, and practical, scripts were carefully developed to bring the world of “SimÉTICO” to life. One such example is presented graphically below:

Figure 5. End of an approved task

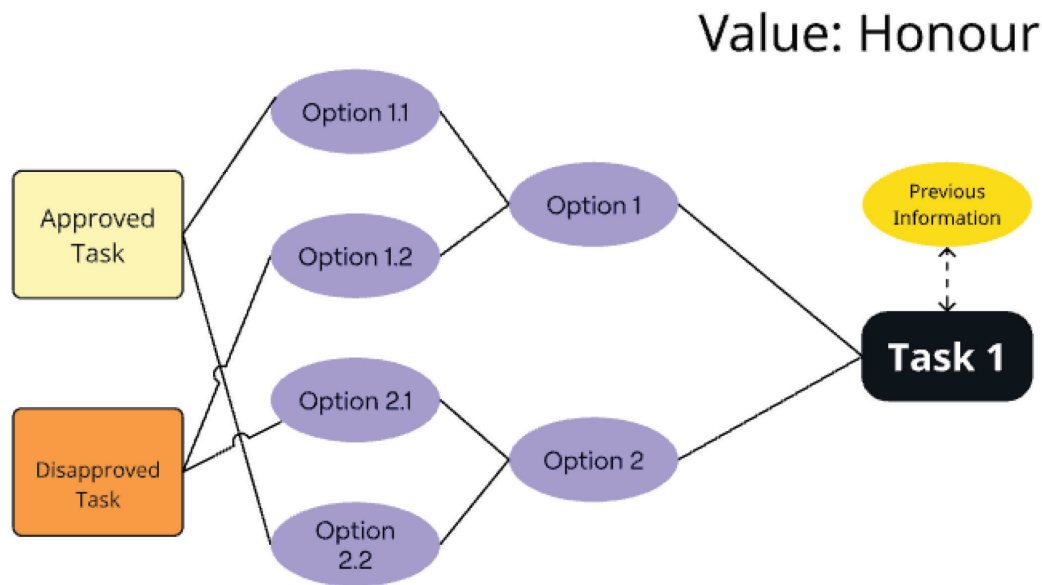


Source: Author.

Figure 6. First situation



Source: Author.

Figure 7. Outline of situations within one of the levels of “SimÉTICO”

Source: Author.

The figure 7 above illustrates the number of scenes and situations included in one of the tasks. However, the process remains consistent across all three missions, with the only variation being the number of possibilities offered in each.

Specifications. Initially, the plan was to create the proposal using 360° videos shot in real military facilities. However, due to security concerns and the necessary permits, this approach was discarded. Instead, two applications were used together to bring “SimÉTICO” to life. The first, Vyond, was employed to generate the animated videos of the proposed scenarios. Vyond allowed for the creation of both main and secondary characters with the required physical attributes and clothing specific to the military context.

The second application, Genially, was used to organize and sequence the previously created videos, structuring the three tasks and linking them to the Code of Military Ethics. It also facilitated the branching mechanism, allowing students’ decisions to shape their journey within the simulation.

However, these two apps, in their free versions, lacked the extensive toolsets necessary for the development of “SimÉTICO.” As a result, the paid monthly versions of both applications were used

for four months—the duration required to build the entire metaverse, considering that the project was developed by a single individual within the constraints of this research.

Additionally, insights from psychology and art were integrated into the design process, particularly regarding color psychology. It’s well known that colors evoke specific emotions—a principle widely applied in fields like marketing. For this project, two colors were selected that align with the proposal’s goals: blue, which is part of the institution’s color scheme, conveys maturity and calmness, while red, in all its shades, symbolizes youth and energy (González, 2019). These color choices were made to reflect the characteristics of both the target population and the institution, aiming to enhance emotional engagement and interaction with the proposal.

Proposal Validation

Thirdly, the validation of the proposal was carried out by a panel of experts, taking into account the available time, the purpose of the project, and the specific context in which it is most relevant. Given the nature of the research, it was essential to engage a group

with expertise in the subject matter and experience in applying such elements to the selected population.

For this validation process, a group of five lecturers were selected. All of them hold a Master's degree

in fields related to education, have at least six years of teaching experience at the selected institution, and expressed interest in participating in the final phase of the research. The following results were obtained:

Table 1. Results of the validation of the proposal

Category	1 Up to date	2 objectives	3 implementations	4 projections	Average
Score	4.9	4.8	5.0	5.0	4.9
Expert	1	2	3	4	Average
1	5	5	5	5	5
2	5	5	5	5	5
3	4.5	4.3	5	5	4.7
4	5	5	5	5	5
5	5	5	5	5	5

Source: Author.

According to the expert consensus, this pedagogical proposal for teaching and learning military ethics effectively meets the objectives outlined in this research. The experts also agree that it is feasible to implement this approach in training processes that can benefit from the use of digital tools, metaverse elements, ICTs, and other resources currently available through the internet.

Finally, the experts recommended expanding the range of proposed situations to encompass a broader variety of scenarios. To achieve this, it would be ideal for the institution to provide support by allocating necessary resources to further strengthen and expand the proposal as soon as possible.

Discussion

The research question guiding this study was: Which didactic strategy could enhance the learning of the military ethics module at the NCO School of the Colombian Aerospace Force? To address this, it is essential to consider the context of the Colombian Air Force NCO School, a prestigious institution known for its strong doctrinal foundation and commitment to achieving high-quality accreditation for its programs. The institution's unique environment is shaped by the collective dedication of

students, faculty, and administrators, all of whom contribute to a distinctive educational experience that is clearly reflected in its graduates.

Currently, the institution's predominant teaching strategy relies on lecture-based classes with a strong focus on theoretical content across all subjects, including military ethics. While instructors are highly experienced in their technical and professional fields, not all have formal training in pedagogy, which presents challenges in implementing cross-cutting approaches that address student motivation, generational differences, and the integration of ICT tools.

A review of the literature provided a solid theoretical foundation, confirming that technology is playing an increasingly significant role in higher education for both teaching and learning. However, there is a noticeable gap in research within military institutions in Colombia, with most contributions in this area coming from North American sources. Moreover, limited research has connected pedagogy, ICT, and ethics in a comprehensive manner, making this study and its proposal particularly innovative.

Initial assumptions about students being "digital natives" were challenged by demographic data

and feedback from students, which revealed that many had limited access to the internet and digital tools before entering the school. This suggests that Prensky's (2014) theory of digital natives does not fully apply in the Colombian context, where connectivity is not universal. However, elements from Howe and Strauss's generational theory, such as the need for motivation and the desire for meaningful learning experiences among millennials and centennials, were evident. As a result, motivation emerged as a crucial factor in engaging students, highlighting the need for preparatory support to ensure successful engagement with new educational approaches.

Ethics was consistently identified as a central theme by both the student population and the literature, underscoring its importance as a catalyst for teaching and its connection to real-life practice through example. Each student's academic program influences how they experience educational processes, with every program maintaining unique characteristics while sharing a common institutional culture. Students expressed a strong sense of identification with both their technical roles and their responsibilities as future military professionals.

A particularly noteworthy finding was the emphasis students placed on learning by example. Throughout the focus group discussions, students highlighted the critical role of instructors and commanders as ethical role models. This aligns with Cortina's (2013) theory that practical experience is essential for internalizing ethical principles. For students, it is vital to see theory reflected in practice and to learn ethical behavior directly from those in leadership positions.

Motivation emerged as another key concern. Students agreed that practical activities were more engaging, easier to understand, and more memorable than purely theoretical lessons. They advocated for the inclusion of applied components alongside theoretical content, both in their training and in their future professional roles. This supports the notion that military ethics education should incorporate practical elements, such as an ethics simulator, which provides a safe and realistic environment for applying theoretical knowledge.

From the faculty's perspective, while the primary goal is to prepare military personnel, the educational process must also address the diverse backgrounds and experiences of the students. The pandemic accelerated the adoption of digital tools, and faculty members now feel better equipped to leverage technologies like the metaverse, virtual reality, and augmented reality to enrich the learning experience—provided there is a clear educational objective.

There is consensus among faculty, experts, and administrators on the need for clear objectives when integrating technological resources into teaching. The work of theorists such as Postman (1975) emphasizes the importance of moving beyond traditional print-based methods to embrace the opportunities offered by technology, preparing students for the challenges of the future. The present proposal aligns with this vision, aiming to provide a motivating, metaverse-based alternative for teaching the elements of the Military Code of Ethics through practical application.

However, the development of such technological solutions is constrained by factors such as time, resources, available personnel, and conceptual clarity. While high-quality technological products can be created, they often require significant investment and time. In resource-limited contexts, simpler solutions must be pursued, though there are existing applications that can facilitate these innovations.

Regarding technology, there are many tools and opportunities for developing new concepts, but free or basic programs often come with significant limitations. Platforms that allow for greater customization, the creation of specific environments, or the selection of avatar features typically require payment, familiarity with the application, and considerable time for development.

The Sim-ETICO architecture directly addresses Arendt's call for judgment as a deliberative, internal process, while simultaneously enabling Habermas' communicative action by requiring the articulation of a justified ethical standpoint. The high expert consensus validates that this design feature is a viable pedagogical mechanism for training ethical reflection.

The findings from the validation process align with the perspectives of theorists such as Postman (1975), who argued that technological change is irreversible; Echeverría (2015), who emphasized the need to adapt technology to academic goals and needs; and Drucker (1993), who advocated for the integration of teaching and learning technologies to support success in contemporary society.

Consequently, this study is considered innovative, relevant, and well-adapted to the context of military ethics education. The proposed Sim-ETICO simulator, similar to those used in pilot training, offers a unique approach to teaching and learning ethics—one for which there is no direct equivalent elsewhere in the world.

Conclusion

In line with the objectives of this research, a diagnostic was conducted within the selected institution, involving students, teachers, and a manager. This collaborative process provided a clear picture of the current situation and helped identify both strengths to maintain and areas for improvement in any alternative approach.

Regarding the second objective, a wide range of resources offered by ICT and the metaverse were explored. However, due to constraints related to personnel, time, and budget, the chosen tools were those best suited to the aims of this research. Applications like Vyond and Genially were selected because they do not require advanced programming skills and allow for the creation of a specific metaverse environment. Nevertheless, it was noted that free versions of such applications have significant limitations in terms of available features.

When the proposal was validated by a panel of experts, valuable feedback was collected on both its structure and content. These recommendations will be instrumental in further strengthening the virtual environment in future iterations of the project.

This research is significant due to its unique, innovative, and creative approach. It has the potential to greatly impact current and future students by providing a user-friendly, accessible resource that fosters growth and adaptation. Additionally, the institution will benefit from better-trained staff in military ethics, as the proposed strategy offers real-life scenarios for decision-making, thereby enhancing the ethical competencies of military personnel.

The next step should be to present the results and the proposal to the educational institution, in order to have its official approval, so that in the medium term, it can be applied with real students, and finally, in the long term, generate a strengthening of the situations and characters created, hand in hand with the resources that the institution can offer.

Finally, it is hoped that this proposal can be used in other educational institutions in which ethics may be of some kind of interest, as well as in other subjects with a high theoretical component, even in other types of contexts and places, considering that this research can be of great contribution to the entire field of knowledge of education. It is clear that the possibility of finding new paths in the educational field of ethics must be studied, analysed, and, why not, applied to understand to what extent and in what way pedagogical innovation can play an important role in the current era, with the characteristics of this generation. It is important to note that the main limitations of this proposal were related to the restricted time available for its development due to the contextual conditions, as well as the inability to conduct a real-world application with experimental and control groups. Additional constraints included limited financial resources, since the project took four months to complete and incurred monthly costs, all managed by a single researcher.

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